

Hierarchical Cross-Parameterization for the Morphing of Deforming Meshes

Lizhou Cao

lc1248@rit.edu

Rochester Institute of Technology
Rochester, New York, USA

Chao Peng

cxpigm@rit.edu

Rochester Institute of Technology
Rochester, New York, USA

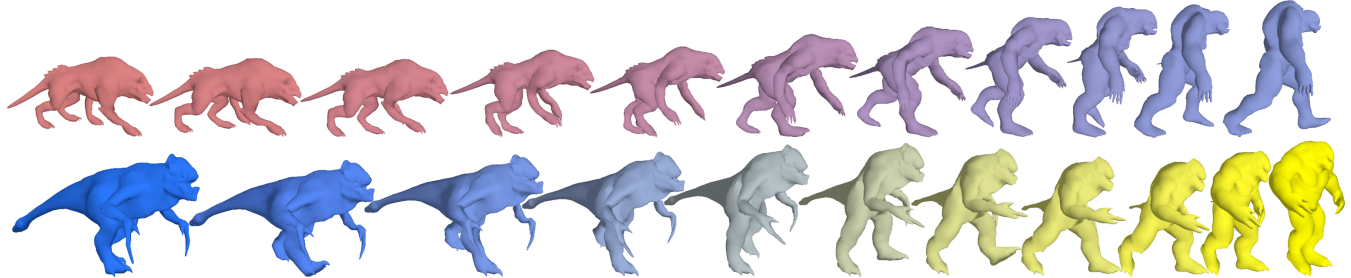


Figure 1: Two morphing sequences generated by the method in this work. The first and last characters in each sequence are the source characters. They are varied in shapes and rigged with different skeletons (bipedal and quadrupeda, tail and no tail).

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1 INTRODUCTION

Deforming meshes with animations are common geometric data used in film, gaming, simulation, digital training, and scientific visualization. A deforming mesh usually consists of two parts: a mesh with topology presenting the shape and a hierarchical set of joints (a skeleton) that deforms mesh regions. Understanding deforming meshes and reusing them to produce new characters is crucial in asset creation. Traditional methods require specific types of input and labor-intensive effort to create character variations in order to enhance the visual diversity in a realistic environment. For example, to create animated and morphologically varied characters for a crowd simulation, sets of interchangeable body parts have to be built manually. The blend-shape technique is an automated method to create morphable mesh sequences, but it requires input meshes must be isomorphic, which means the input meshes must have the same number of vertices and the same topology with a one-to-one mapping of all vertices.

The goal to automate the generation of morphologically varied characters from non-isomorphic inputs has led to the use of *cross-parameterization* techniques [Kwok et al. 2011; Peng and Timalsena 2016]. Here, cross-parameterization is able to form a one-to-one vertex mapping for input meshes into a common representation on

a parametric domain. Feature correspondences can be identified and aligned in the parametric domain. However, most existing methods only consider static meshes and find feature correspondences solely based on surface properties [Hu et al. 2017; Peng and Timalsena 2016; Praun and Hoppe 2003]. When finding feature correspondences for deforming meshes, deformation properties like joints and skinning weights (influences from different joints) should be considered, which yet are not well studied in existing methods.

To automate the creation of morphologically varied characters with animations, this work presents a hierarchical spherical cross-parameterization method for the morphing of deforming meshes. Figure 1 shows two morphing sequences generated by our method. Our method takes animated characters as input and is able to generate morphologically varied animated characters. Deforming features are aligned automatically in a hierarchical spherical domain, which ensures the synthesized character retains meaningful changes in the shape in response to the update of animations. Our cross-parameterization method first segments the mesh based on deforming features, and then maps the segments to spheres through an iterative leaf-to-root spherical parameterization. The alignment employs an optimization process that considers the match of deforming feature correspondence between parent spheres and child spheres. In advancing the existing parameterization methods, our method significantly mitigates the distortion rates at feature-intensive regions in the parametric domain (Figure 2). Our alignment method conveys semantic matching for deforming feature correspondence, which is more suitable for deforming meshes.

2 APPROACH

Segmentation based on deforming features. Skinning weights are a common numerical presentation of deformation that can be created procedurally or manually. A skinning weight defines the percentage of influence from a joint on a vertex. Gradients of skinning weights can be used to form *boundaries* between parent and child parts [Lee et al. 2006], as shown in Figure 3. These boundaries are considered the major deforming features in our method.

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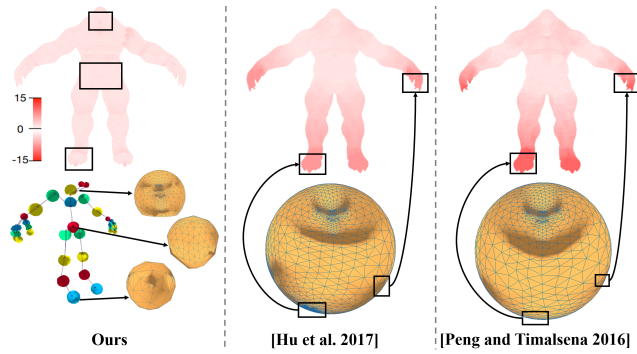


Figure 2: The comparison of area distortions mapped on the 3D model. Distortions are colored to triangles.

We employed a part-based segmentation method to segment the mesh based on the boundaries formed from skinning weights, where each segment should be matched to a joint. To do this, we labeled each vertex with the joint index that gives the highest skinning weight value. Vertices with the same label form a segment. It is possible that non-adjacent regions are labeled with the same joint index, such that more than one of the segments are matched to the same joint. This would cause a parent-child mismatch in the later building of spherical representations. Also, it is possible that two adjacent vertices are labeled with non-adjacent joint indices, which would cause adjacent regions to be treated as non-adjacent segments. To solve these issues of labeling, we developed a label refinement method. It can merge the separate regions that are labeled with the same joint index to the nearest ancestor joint, in which they can connect through a set of adjacent vertices. It rechecks the labels of adjacent vertices and ensures they are labeled with adjacent joint indices.

Building hierarchical spherical representations. For each segment, we implemented a vertex decimation method [Hu et al. 2017] to collapse the segment down to a base shape and project it onto a sphere. We created the spheres in a hierarchical manner by starting the parameterization with the segments associated to leaf joints. A leaf segment contains only one *parent boundary*, which connects to the corresponding *child boundary* in its parent segment (Figure 3). Non-boundary vertices of the leaf segment are decimated to a single vertex that we call *representative vertex*, denoted as $\mathbf{rv}$. As a result, the base shape of the leaf segment is created with the $\mathbf{rv}$ connecting to the parent boundary vertices, which is a pyramid shape with an open bottom. The $\mathbf{rv}$ is duplicated to the parent segment and connected with every vertex of the corresponding child boundary. With such leaf-to-root parameterization process, child boundaries in each segment can be filled with the $\mathbf{rv}$ s. Vertices on the filled child boundaries are decimated in the same way as regular vertices. For the root segment that does not have a parent, its base shape is a solid tetrahedron. Each base shape is projected to an individual sphere. After that, the rest of the vertices are inserted back one by one and projected to the locally least-distorted location inside the one-ring region.

Two-step feature alignment. We conducted a two-step alignment to coincide deforming correspondences between sphere hierarchies. In the first step, we applied a rotation matrix for one sphere to align with the second sphere. To calculate the rotation matrix, we

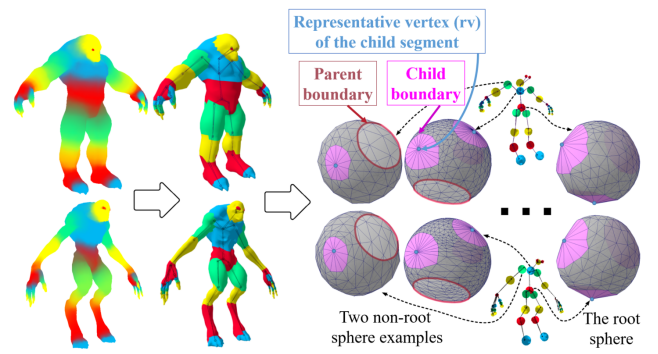


Figure 3: A hierarchical parameterization and alignment with parent and child boundary features. The input rigged characters are segmented based on skinning weights. Segments are parameterized to spheres in accordance with the skeletal structure. The boundaries from the segmentation are deforming features for aligning pairs of corresponding spheres.

minimized the sum of deformation distances referenced to the child boundary in the parent. Since the root sphere does not have parent boundaries, the deformation distance for the root sphere is defined as the sum of spherical distances between every pair of corresponding $\mathbf{rv}$ s in the child boundaries. Given two corresponding non-root spheres, we first rotated the spheres to ensure the parent boundaries were in the same plane; and then we rotated the first sphere about the axis perpendicular to the plane section of the parameterized parent boundary, so that child boundaries between two corresponding spheres can be brought near each other as closely as possible. In the second step, child boundaries are aligned to exact positions, and the rest of the vertices are interpolated with a radial basis function [De Boer et al. 2007].

Remeshing and morphing. A 3D morphing application is an appropriate way to evaluate if the aligned features in a parametric domain convey semantic matching between shapes in 3D space. To create morphing, we combined and remeshed all pairs of spheres by intersecting their edges, and then an adaptive isotropic mesh was generated in regards to the distribution of all input vertices and intersection points [Alliez et al. 2003]. We plan to evaluate the quality of our cross-parameterization method by processing more characters with morphological variations and different rigs.

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